PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

MIDTERM EXAM
10/28/25

Name _____
NetID _____

BEFORE YOU START…

On the exam, blue book, and answer sheet, write:
- your FULL name (as listed by the university registrar)
- your NetID (i.e. your NYU email minus "@nyu.edu")

1. The gauge pressure in each of the four tires of a car is 260 kPa. If each tire has a "footprint" of 190 cm^2, (area touching the ground), what is the mass of the car?

a) 1.59×10^3 kg b) 2.02×10^3 kg c) 1.71×10^3 kg
d) 1.86×10^3 kg e) 2.17×10^3 kg

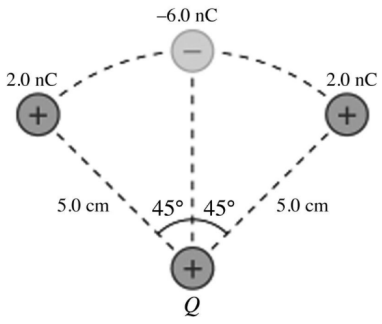
2. Water flows through a 5 cm diameter horizontal pipe that gradually narrows to a diameter of 4.5 cm. In the wider part of the pipe, the gauge pressure is 35 kPa and the water flows at a speed of 2.5 m/s. What is the gauge pressure in the narrow part of the pipe?

a) 19.8 kPa b) 6.91 kPa c) 2.37 kPa d) 33.4 kPa e) 28.2 kPa

3. A cube with sides of length 10 cm made of an unknown material floats at the surface between water and oil. The oil has a density of 800 kg/m^3. If the cube floats so that it is 72% in the water and 28% in the oil, what is the mass of the cube?

a) 0.944 kg b) 0.955 kg c) 0.966 kg d) 0.933 kg e) 0.922 kg

4. The point charge at the bottom of the figure is Q = 16 nC. Each of the two upper side charges is +2.0 nC, and the top charge is −6.0 nC. All three charges lie on a circular arc of radius r = 5.0 cm centered on Q. The magnitude of the net electrostatic force on Q due to the three other charges is closest to:

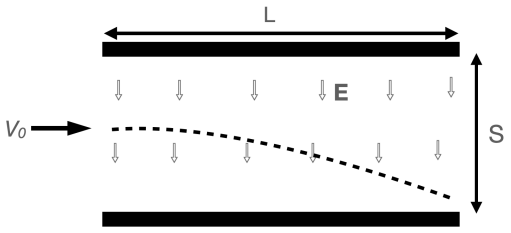


a) 1.63×10^{-4} N b) 2.17×10^{-4} N
c) 1.94×10^{-4} N d) 1.82×10^{-4} N
e) 2.05×10^{-4} N

5. An atomic nucleus has a charge of +41e. The magnitude of the electric field at a distance of 1.0 m from the center of the nucleus is closest to:

a) 5.90×10^{-8} N/C b) 6.76×10^{-8} N/C c) 6.19×10^{-8} N/C
d) 5.32×10^{-8} N/C e) 6.47×10^{-8} N/C

6. A proton is projected horizontally, with speed v_o, midway between two parallel plates that are separated by S = 0.50 cm. The electric field between the plates has magnitude E = 6.10 × 10^4 N/C, directed vertically downward. The plates are L = 5.60 cm long. Find v_o if the proton just misses the lower plate as it emerges from the field.

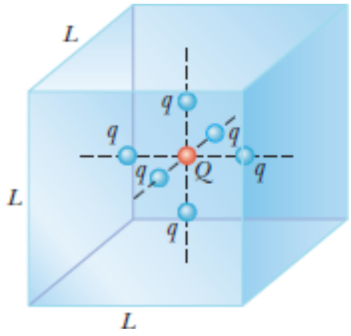


a) 1.91 x 10^6 m/s b) 1.62 x 10^6 m/s c) 1.75 x 10^6 m/s
d) 2.98 x 10^6 m/s e) 1.51 x 10^6 m/s

7. A very large thin sheet in the xy plane carries a uniform charge per unit area $+\boldsymbol{\sigma}$. A second large thin sheet has a charge per unit area $-17\boldsymbol{\sigma}$ is located parallel to the first sheet at z = +2d. Far from the edges of the sheets, what is the magnitude of the net electric field at a point where z = +2.5d?

a) 5 $\boldsymbol{\sigma}/\epsilon_0$ b) 6 $\boldsymbol{\sigma}/\epsilon_0$ c) 7 $\boldsymbol{\sigma}/\epsilon_0$ d) 8 $\boldsymbol{\sigma}/\epsilon_0$ e) 9 $\boldsymbol{\sigma}/\epsilon_0$

8. A particle with positive charge $+\boldsymbol{Q}$ is located at the center of a cube of edge length *L*. In addition, six other identical particles each with charge $\boldsymbol{q}$ = -4.5$\boldsymbol{Q}$ are positioned symmetrically around the central charge, as shown. What is the electric flux through *only the bottom face* of the cube?



a) -(16/3) $\boldsymbol{Q}/\epsilon_0$ b) -(13/3) $\boldsymbol{Q}/\epsilon_0$ c) -(10/3) $\boldsymbol{Q}/\epsilon_0$
d) -(7/3) $\boldsymbol{Q}/\epsilon_0$ e) 0

9. If a point charge of magnitude 2.0 μC is fixed in place at the origin, what is the work required to move a point charge of 6 μC from the point on the x-axis where x = 5 m to the point on the y-axis where y = 3 m?

a) 71.9 mJ b) 36.0 mJ c) 57.5 mJ d) 18.4 mJ e) 14.4 mJ

10. A conducting sphere with radius of 10.0 cm is charged to 450 V (assuming that V = 0 as r goes to infinity). What is the surface charge density on the sphere?

a) 7.97 nC/m^2 b) 19.9 nC/m^2 c) 39.8 nC/m^2
d) 26.6 nC/m^2 e) 13.3 nC/m^2